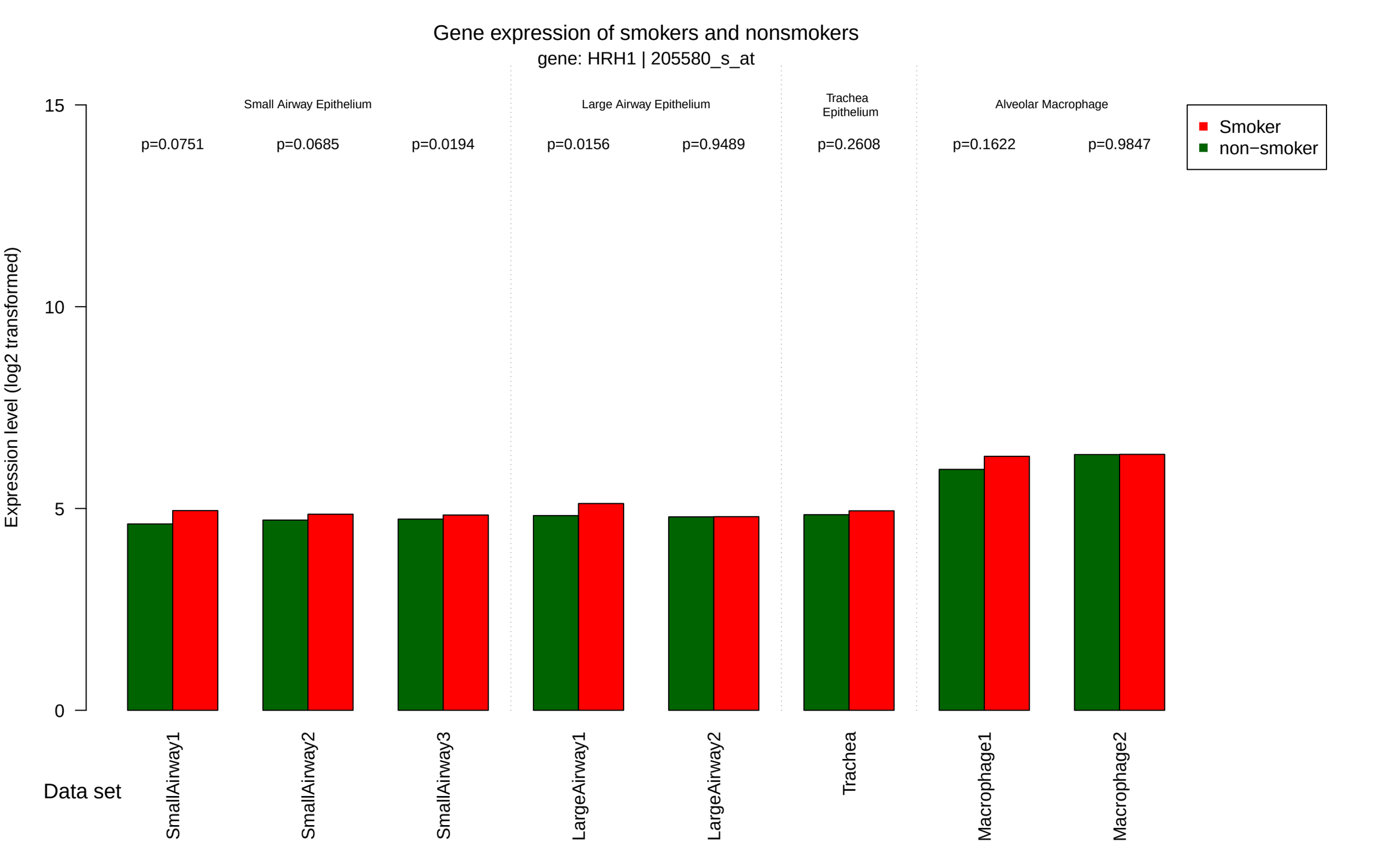
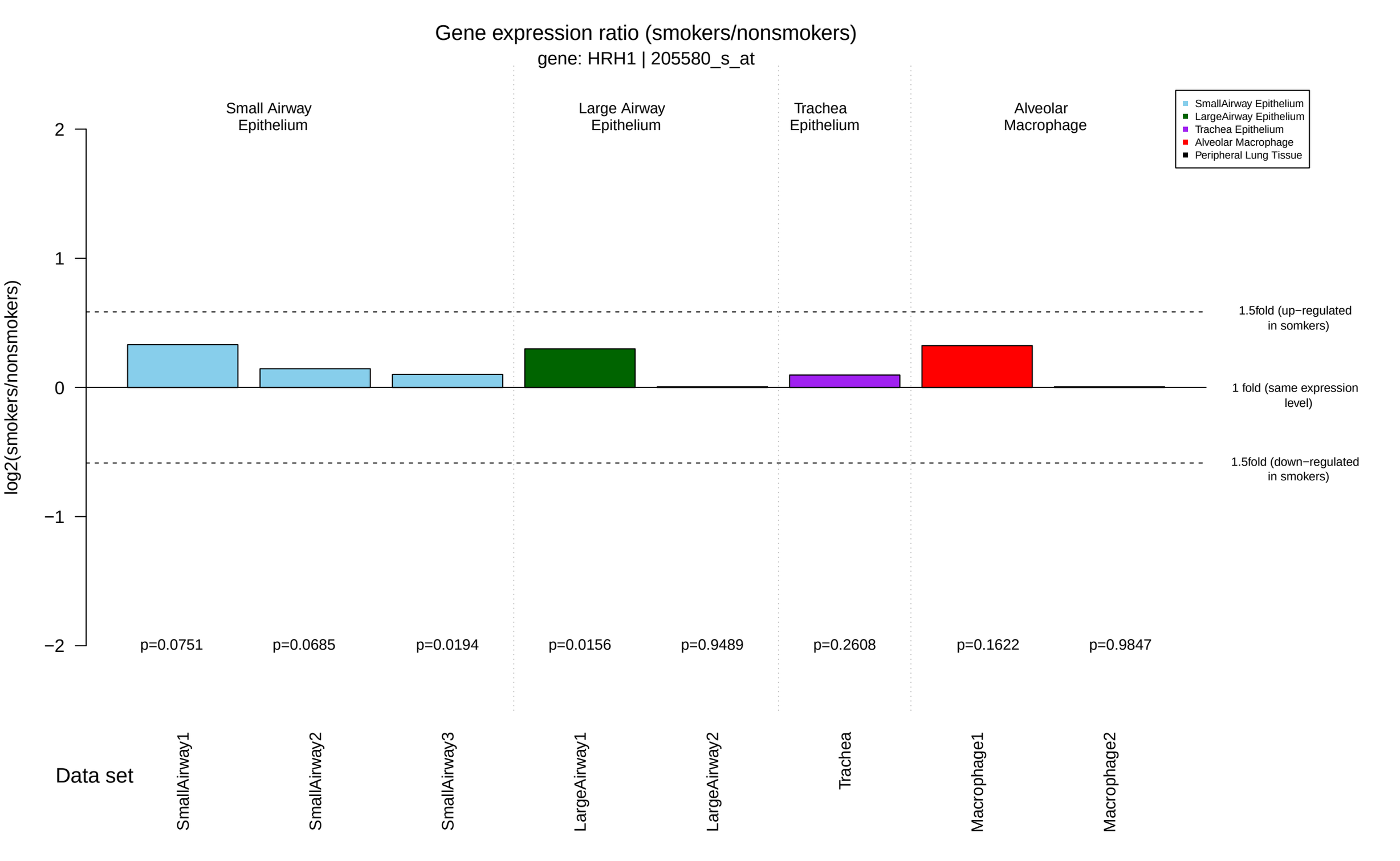
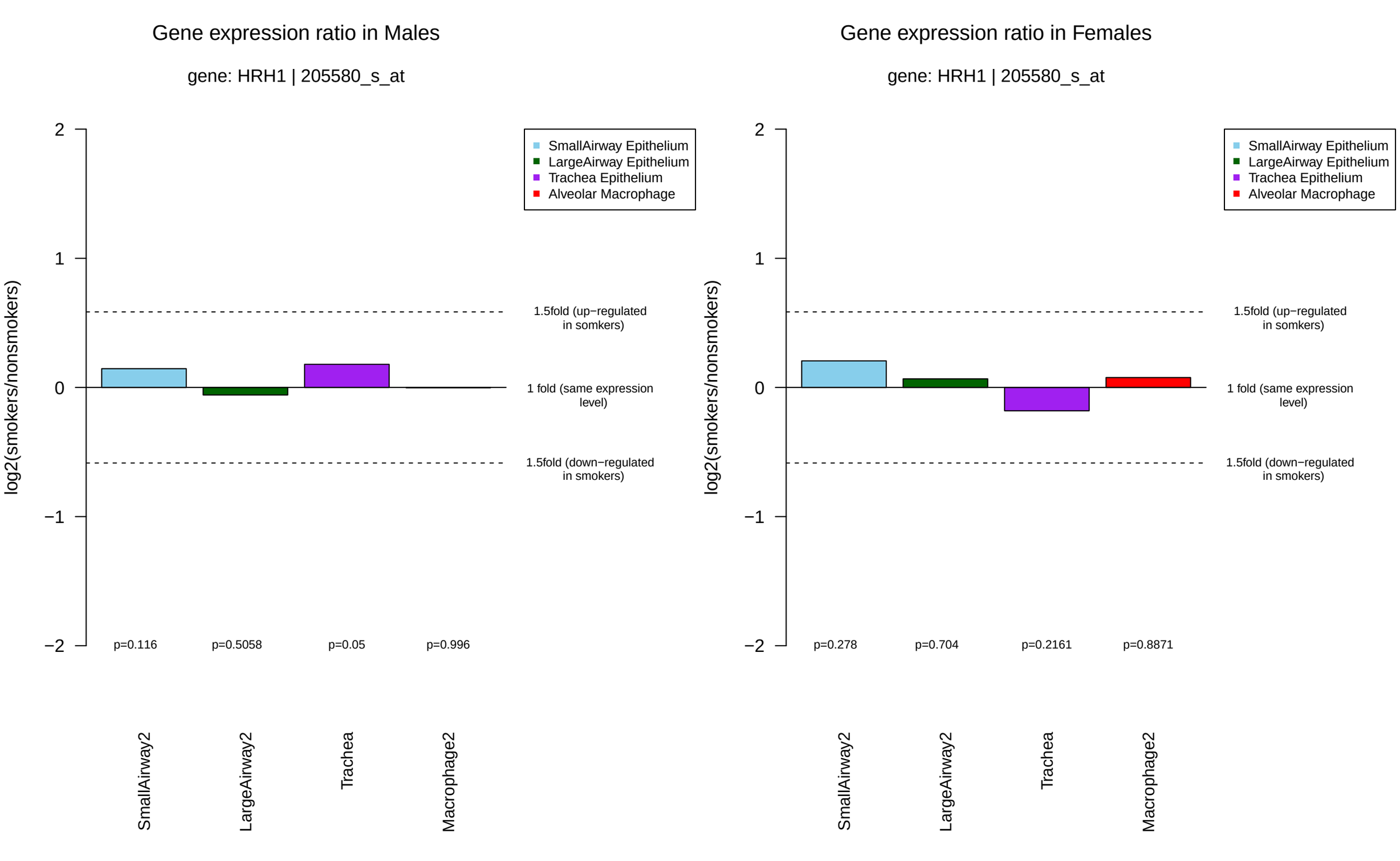
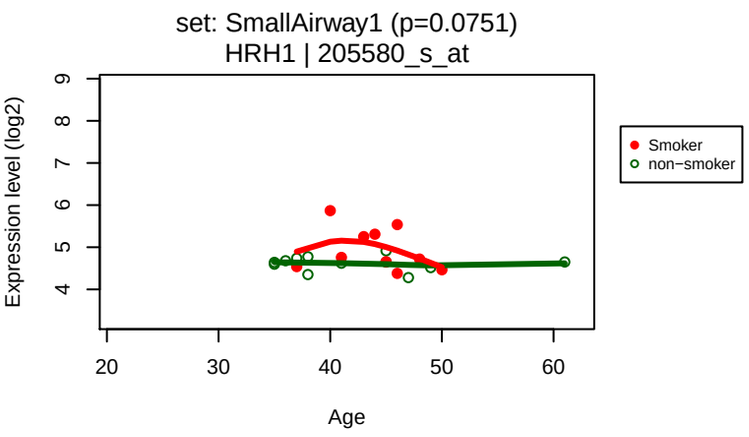
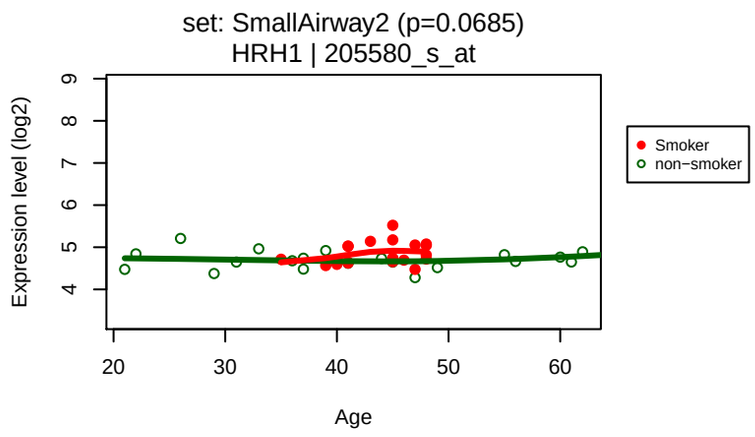
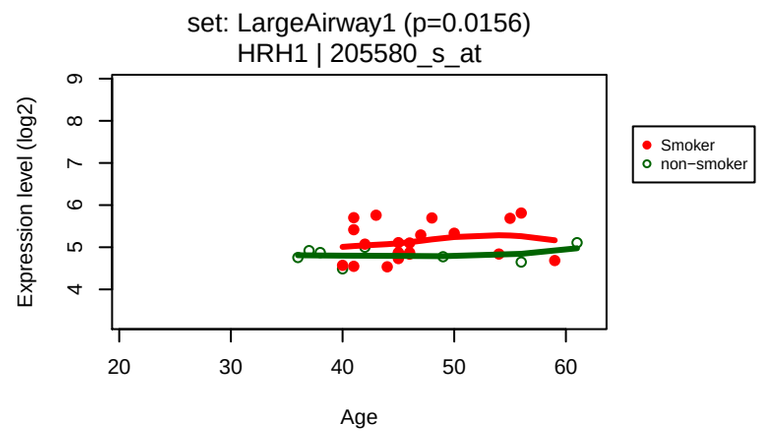

Gene expression of smokers and nonsmokers
gene: HRH1 | 205580_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0751
p=0.0685
p=0.0194
p=0.0156
p=0.9489
p=0.2608
p=0.1622
p=0.9847
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

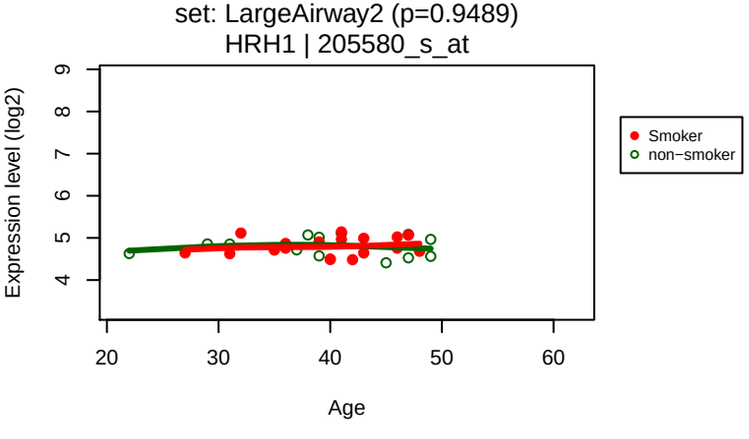

Gene expression ratio (smokers/nonsmokers)
gene: HRH1 | 205580_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0751
p=0.0685
p=0.0194
p=0.0156
p=0.9489
p=0.2608
p=0.1622
p=0.9847
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

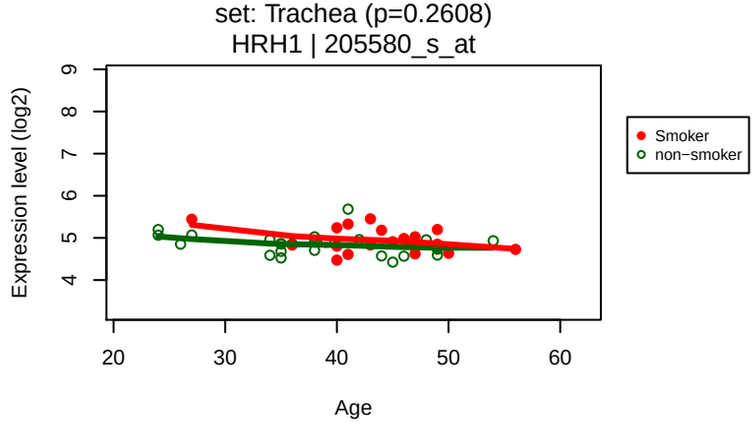

Gene expression ratio in Males
gene: HRH1 | 205580_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.116
p=0.5058
p=0.05
p=0.996
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HRH1 | 205580_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.278
p=0.704
p=0.2161
p=0.8871
SmallAirway2
LargeAirway2
Trachea
Macrophage2

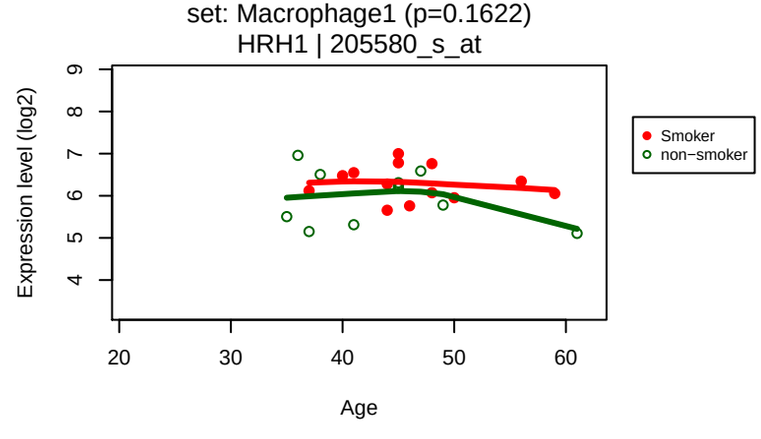

set: SmallAirway1 (p=0.0751)
HRH1 | 205580_s_at
Expression level (log2)
Age
Smoker
non−smoker

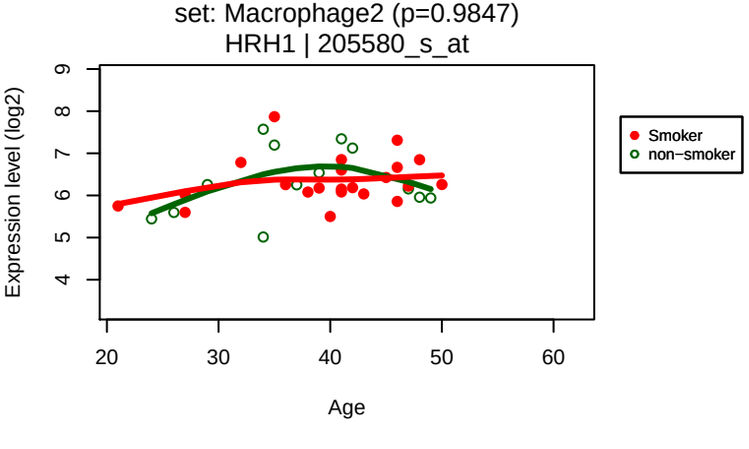

set: SmallAirway2 (p=0.0685)
HRH1 | 205580_s_at
Expression level (log2)
Age
Smoker
non−smoker

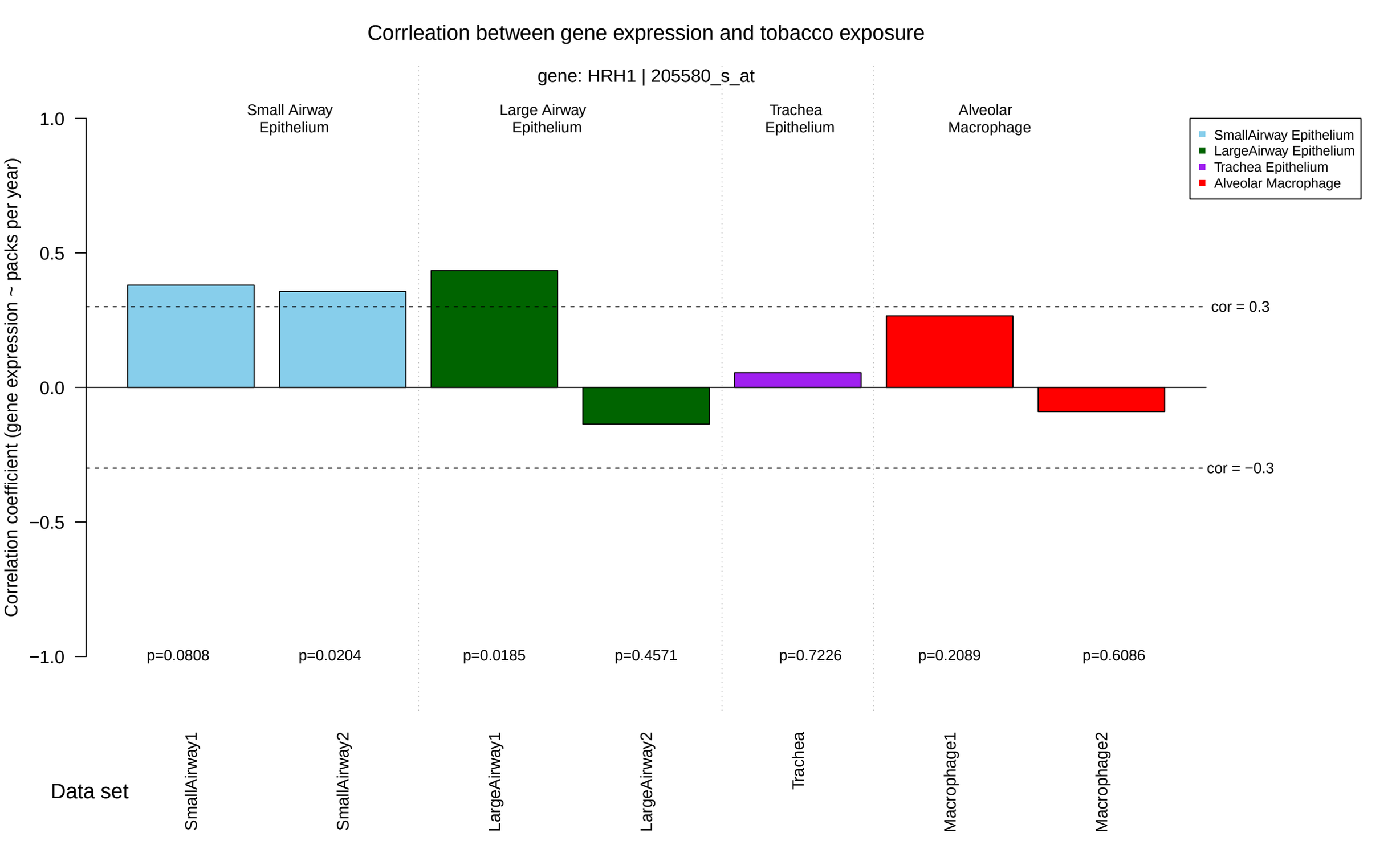

set: LargeAirway1 (p=0.0156)
HRH1 | 205580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9489)
HRH1 | 205580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2608)
HRH1 | 205580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1622)
HRH1 | 205580_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9847)
HRH1 | 205580_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HRH1 | 205580_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0808
p=0.0204
p=0.0185
p=0.4571
p=0.7226
p=0.2089
p=0.6086
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2